

Table 2: TCP Connect scan

Command and results	Explanation
<code>nmap -PN -sT -vv -n -p1-1000 -T4 -oNmapTCPConnect.txt 117.X.X.X</code>	-n does not do reverse DNS, thus saving time; -T4 is for faster execution of the command.
Starting Nmap 5.21 (http://nmap.org) at 2011-01-23 15:23 IST	
Initiating Connect Scan at 15:23	
Scanning 117.195.44.148 [1000 ports]	
Connect Scan timing: About 30.00% done; ETC: 15:25 (0:01:12 remaining)	
Connect Scan timing: About 60.00% done; ETC: 15:25 (0:00:41 remaining)	
Completed Connect Scan at 15:25, 101.04s elapsed (1000 total ports)	
Nmap scan report for 117.X.X.X	
Host is up.	
All 1000 scanned ports on 117.X.X.X are filtered	First 1000 TCP ports found closed.

TCP Connect Scan, as well as the SYN scan, finds all ports between 1 and 1000 as filtered ports. External access stands blocked. Let us continue scanning the firewall for ports between 1001 to 65535 (Table 3). To save scan time, reduce the RTT timeout.

Table 3: SYN scan of TCP ports 1001 to 65535

Command and results	Explanation
<code>nmap -PN -sS -f -w -n -p1001-65535 -T4 -max-rtt-timeout 15 -oNmapSYNScan.txt 117.X.X.X</code>	Timing settings changed for better performance
WARNING: You specified a round-trip time timeout (15 ms) that is EXTRAORDINARILY SMALL. Accuracy may suffer.	As soon as the timing is changed to a very low value, Nmap warns of a possible problem with its accuracy.
Nmap scan report for 117.X.X.X	

Host is up	
All 64535 scanned ports on 117.195.44.128 are filtered	All ports are filtered—none are open.
Nmap done: 1 IP address (1 host up) scanned in 209.93 seconds	Make a note of the time required. To scan 1000 ports, it required about 101 seconds in the earlier scan.

Continue probing further to detect open UDP ports (Table 4).

Table 4: UDP ports scan

Command and results	Explanation
<code>nmap -PN -sU -vv -n -p U:1-65535 -T4 -max-rtt-timeout 15 -oNmap-SYNScan.txt 117.X.X.X</code>	UDP port scan, scanning all ports
Starting Nmap 5.21 (http://nmap.org) at 2011-01-23 15:47 IST	
Initiating UDP Scan at 15:47	
Completed UDP Scan at 15:50, 212.71s elapsed (65535 total ports)	
Nmap scan report for 117.195.44.128	
Host is up.	
All 65535 scanned ports on 117.195.44.128 are open/filtered	All ports are filtered—none are open.
Nmap done: 1 IP address (1 host up) scanned in 212.94 seconds	

After scanning all 65535 TCP and UDP ports, we find that none of them are open. Now, the firewall scanning continues by scanning it internally. Here, the command used is:

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nmap -vv -sU -sT -p1-1000 -n -r -T4 -oNmapIPCopInternal.txt 192.168.1.1
```

Starting Nmap 5.21 (http://nmap.org) at 2011-01-25 21:56 IST	Nmap scan starts; various scanning techniques used, as detailed in the command line.
Completed ARP Ping Scan at 21:56, 0.00s elapsed (1 total hosts)	
Completed UDP Scan at 21:57, 23.10s elapsed (1000 total ports)	